Pratik Mehta

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Experience

Senior AI Engineer Nov 2025 – Present
Xpergia *Mumbai, India*

- Co-developed an accessibility application to scan HTML pages/PDF files for violations of WCAG accessibility criteria, and remediate discovered violations. No tool in the market offered remediation of PDF accessibility issues at the time
- Led system design and development for an LLM-based statistics and insights generation system based on live football match data for a sports / media client

Consultant Nov 2020 – Jun 2024
Deloitte Digital USI *Mumbai, India*

- Designed & developed a microservice-based solution containing REST APIs served over AWS to answer user questions about live operations KPI data in a chatbot setting for a large pharmaceutical enterprise client
- Developed a proof-of-concept application that takes as input pictures of walls, and returns the closest color from a paint company's catalog
- Conducted org-wide deep learning training sessions as part of AI/ML team's Democratize AI initiative

Research Assistant Intern Jun 2018 – Sept 2018
Information Sciences Institute, University of Southern California *Waltham, MA*

- Studied correlation between different evaluation metrics for machine-translation in cross-lingual information retrieval (CLIR) systems

Research Engineer Jun 2014 – Jul 2016
Indian Institute of Technology Bombay *Mumbai, India*

- Designed an error analysis engine to identify and classify translation errors made by the Shata-Anuvadak (100 translators) Statistical Machine Translation (SMT) system developed at IIT Bombay
- Performed oracle experiments to confirm that simply changing the word-order of automatic translations between English-Indian language Statistical Machine Translation (SMT) systems could improve BLEU scores by as much as 60.45%.
- Trained a (then) state of the art morphological analyzer for Marathi
- Trained and deployed phrase-based statistical machine translation systems for English-Marathi and English-Indonesian using medical-domain textbooks

Education

University of Massachusetts Amherst Amherst, MA
Master of Science in Computer Science, ***GPA****: 3.21* *Aug 2016 – May 2019*

D. J. Sanghvi College of Engineering, University of Mumbai Mumbai, India
Bachelor of Engineering, ***First class*** *Aug 2011 – Jun 2015*

Publications

The challenges of optimizing machine translation for low-resource CLIR | *EMNLP '19*
Constantine Lignos, Daniel Cohen, Yen-Chieh Lien, Pratik Mehta, W Bruce Croft, Scott Miller

The IIT Bombay English-Hindi parallel corpus | *LREC '18*
Anoop Kunchukuttan, Pratik Mehta, Pushpak Bhattacharyya

Investigating the potential of post-ordering SMT output to improve translation quality | *ICON '15*
Pratik Mehta, Anoop Kunchukuttan, Pushpak Bhattacharyya

Technical Skills

Languages: Python (main), SQL (intermediate), Rust (basic),
Frameworks/Libraries: FastAPI, streamlit, pytorch, pandas, numpy, scikit-learn, langchain
Developer Tools: Git, Docker, Copilot, Claude Code
Cloud Platforms: AWS, Azure

Projects

Weak lensing ML uncertainty challenge | *pytorch w/ Gemini Pro* Jan 2024

- Used normalizing flows to address an OoD problem: Finding whether actual 2-D convergence maps (images of deep space featuring weak gravitational lensing) differ from simulated ones

Alfred: LLM powered hyper-personalized notifications | *Llama* Jan 2024

- Developed a proof of concept simulator to track a web app users' English-language notification preferences based on simulated live-data events on the application

LegalEase: Cross-lingual semantic search engine | *Cohere, Qdrant* Mar 2023

- Developed an application (now called RAG) that answers questions factually based on lengthy legal documents as well as a multi-lingual semantic search engine for any-language-query to any-language-document retrieval
- Was a finalist in the Semantic Search hackathon co-organised by lablab.ai

Semantic Segmentation in cardiac MRI images | *Keras, Tensorflow* Dec 2016

- Developed a CNN-based model for semantic segmentation to find the boundary of the right ventricle of the heart in cardiac MRI images to estimate ejection fraction rate and predict cardiomyopathy